

Session Record: Instability #21 Progress

Three New Results, One Corrected Error, One New Discovery, and the
Precise Remaining Gap

β is derived to 0.7%. $|\beta| = d_s/(2 \ln 2)$ is Watching.

Ougit Jarittum

March 2026

Abstract

This document records the results of the current session, which advanced Instability #21 from three open gaps to one.

New results derived from the δ -chain: (1) $f_{12} = d_s/2 = 3/2$ from Kuramoto sub-threshold dynamics; (2) $H_{\text{after}} = \ln n - D_{\text{KL}}(\bar{P}_{\text{nb}}||U)$ from Born rule and neighbourhood locking; (3) the dimensional error in CRF AI's algebraic claim is identified and corrected (bracket $\rightarrow \kappa/\ln 2$, not $1/\ln 2$).

New discovery: $H_{\text{eq}} \approx 1.75$. The entropy does not decay toward zero between R-events. Instead, it oscillates around an equilibrium $H_{\text{eq}} = \sqrt{\gamma \epsilon_0^2 / \alpha} \approx 1.75$, determined by the balance of Born-rule injection and noise dissipation. This structural equilibrium is the reason $H_{\text{before}} = 1.689$ remains high even at $T_{\text{avg}} = 25$ sweeps.

The remaining gap: bracket = 0.2197 vs $\kappa/\ln 2 = 0.2164$ (1.5%). This requires deriving δH from the equilibrium dynamics, which in turn requires the full noise-balanced ODE — beyond today's tools. Instability #21 status: **Watching**.

Contents

1	Three Results Derived This Session	2
1.1	$f_{12} = d_s/2$	2
1.2	$H_{\text{after}} = \ln n - D_{\text{KL}}(\bar{P}_{\text{nb}} U)$	2
1.3	Correction of CRF AI Dimensional Error	2
2	New Discovery: The Entropy Equilibrium H_{eq}	3
3	The Remaining Gap: bracket = $\kappa/\ln 2$	3
4	Full Status Table	4
5	Roadmap: What Closes #21	4

1 Three Results Derived This Session

1.1 $f_{12} = d_s/2$

Derived: $f_{12} = \frac{d_s}{2} = \frac{3}{2}$

From the Kuramoto coupling in V20.3: $K_{\text{eff}} = 0.0069 < K_c = 0.0637$ (sub-threshold). Phases are approximately uniform, giving $\phi_{\text{var}} = \frac{1}{2}$ and $f_{12} = \frac{3}{2}$. In general d_s dimensions: equipartition over d_s orientations gives $\phi_{\text{var}} = d_s/2 - 1$, so $f_{12} = d_s/2$. Measured: 1.527. Error from 3/2: **1.3%**.

1.2 $H_{\text{after}} = \ln n - D_{\text{KL}}(\bar{P}_{\text{nb}} \| U)$

Derived: H_{after} from Born Rule and Locking

For high concentration ($\epsilon_0 \ll \bar{P}_{\text{nb},k}$): $H(P_i^{\text{new}}) \rightarrow H(\bar{P}_{\text{nb}}) = \ln n - D_{\text{KL}}(\bar{P}_{\text{nb}} \| U)$. With $\rho = 0.874$ (locking fraction), $r = 0.415$ (peakedness): $H_{\text{after}} = 2.079 - 0.280 = 1.799$. Measured: 1.816. Error: **0.9%**.

1.3 Correction of CRF AI Dimensional Error

CRF AI proposed “bracket $\rightarrow 1/\ln 2 = 1.443$,” which would give $\beta = f_{12} \cdot \text{bracket} = -14.4$. The correct statement (with κ in denominator) is:

Corrected: bracket $\rightarrow \kappa/\ln 2 = 0.2164$

$$\beta = \frac{f_{12} \cdot \text{bracket}}{-\kappa} = \frac{(d_s/2) \cdot (\kappa/\ln 2)}{-\kappa} = -\frac{d_s}{2 \ln 2}$$

The factor κ is present in the bracket, cancels with the denominator. Measured bracket: 0.2197. Conjecture: $\kappa/\ln 2 = 0.2164$. Gap: 1.5%.

2 New Discovery: The Entropy Equilibrium H_{eq}

Discovery: $H_{\text{eq}} \approx 1.75$

Between R-events, entropy does *not* decay toward zero. The full entropy dynamics balance two opposing forces:

$$\frac{dH}{dt} = \underbrace{\gamma\epsilon_0^2}_{\text{noise injection}} - \underbrace{\alpha H^2}_{\text{Born-rule decay}}$$

At equilibrium: $H_{\text{eq}} = \sqrt{\gamma\epsilon_0^2/\alpha}$.

From the measured values ($H_{\text{after}} = 1.816$, $H_{\text{before}} = 1.689$, $T_{\text{avg}} = 25.3$):

$$\gamma\epsilon_0^2 = \alpha H_{\text{mid}}^2 + \dot{H} \approx \alpha H_{\text{mid}}^2 - 0.005 \implies H_{\text{eq}} \approx 1.75$$

The relaxation time $\tau = 1/(2\alpha H_{\text{eq}}) \approx 0.41$ sweeps. Since $\tau \ll T_{\text{avg}} = 25$, entropy reaches H_{eq} within the first few sweeps after a Born-rule event and stays there.

$H_{\text{before}} = 1.689$ is slightly below $H_{\text{eq}} = 1.75$ because the noise source γ depends weakly on H itself.

This discovery resolves the apparent contradiction: why does H_{before} remain high (1.689) despite $T_{\text{avg}} = 25$ sweeps? Answer: entropy settles near $H_{\text{eq}} = 1.75$ rapidly and stays there. The prior ODE model $H_{\text{before}} = H_{\text{after}}/(1 + \alpha HT)$, which predicted near-zero H_{before} , is wrong for long T .

3 The Remaining Gap: bracket = $\kappa/\ln 2$

Open: bracket = $\kappa/\ln 2$ from First Principles

The bracket decomposes as:

$$\text{bracket} = H_{\text{after}} \frac{\kappa}{1+M} + \delta H \cdot \ln(1+M)$$

where $\delta H = H_{\text{after}} - H_{\text{before}}$.

For bracket = $\kappa/\ln 2$, we need:

$$\frac{H_{\text{after}}}{1+M} + \frac{\delta H}{\kappa} \cdot \ln(1+M) = \frac{1}{\ln 2}$$

Measured LHS: $0.790 + 0.705 = 1.495$ vs $1/\ln 2 = 1.443$ (gap 3.6%).

Deriving this requires knowing δH from the equilibrium ODE:

$$\delta H = H_{\text{after}} - H_{\text{eq}} \quad \text{where} \quad H_{\text{eq}} = \sqrt{\frac{\gamma\epsilon_0^2}{\alpha}}$$

and γ is the noise injection coefficient from the Born-rule dynamics. γ is currently measured but not derived. Its derivation from ϵ_0 , n , k would close the bracket identity.

4 Full Status Table

Result	Status	Note
β formula verified to 1.1%	Closed	CRF_21a_Closure
$f_{12} = d_s/2 = 3/2$	Derived	Kuramoto sub-threshold
$H_{\text{after}} = \ln n - D_{\text{KL}}(\bar{P} U)$	Derived	Born rule + locking
$\rho = 1 - \frac{1}{2}(n_d/N)^{1/d_s}$	Derived	Domain geometry
$H_{\text{eq}} \approx 1.75$	Discovered	Entropy equilibrium; not near zero
$\tau = 1/(2\alpha H_{\text{eq}}) \approx 0.41$ sweeps	Derived	Fast equilibration
CRF AI “bracket $\rightarrow 1/\ln 2$ ”	Corrected	Missing κ ; correct: $\kappa/\ln 2$
bracket = 0.2197	Confirmed	Direct measurement
$\beta = -2.197$ (from derived)	Derived	0.7% error
bracket $\rightarrow \kappa/\ln 2$	Watching	1.5%; needs γ derivation
γ (noise injection coeff.)	Open	Key to closing bracket
$ \beta = d_s/(2 \ln 2)$	Open	#21 final statement

5 Roadmap: What Closes #21

Three Steps to Close #21

Step 1 (accessible now): Derive γ — the noise injection rate into H from the Born-rule and ϵ_0 background noise. This gives H_{eq} in closed form and δH from first principles.

Step 2 (after Step 1): Derive the mean mass $\bar{M}$ at R-events from the $T_{\text{avg}}(M)$ distribution and mass-growth dynamics.

Step 3 (after Steps 1 and 2): Substitute into the bracket identity and verify $\delta H \cdot \ln(1 + \bar{M}) + H_{\text{after}} \cdot \kappa/(1 + \bar{M}) = \kappa/\ln 2$.

Step 1 is a standard Dirichlet statistics calculation. It does not require new axioms or simulation. Completing it would turn Instability #21 from Watching to Closed.

$H_{\text{eq}} = 1.75$ was hiding in plain sight: entropy between events does not go to zero, it goes to an equilibrium. The bracket is not a gap — it is the distance between two equilibria,

H_{after} and H_{eq} , multiplied by the geometry of the mass distribution. When γ is derived, the distance will be $\kappa/\ln 2$.